StrayCare: An AI-Driven Full-Stack Framework for Intelligent Stray Animal Rescue, Tracking, and Adoption Optimization

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***Abstract*—The escalating stray animal population presents a significant challenge to municipal management, public health, and animal welfare. Existing digital platforms for rescue and adoption—while automating specific elements—often operate as isolated systems lacking a unified ecosystem for synchronized oversight, traceability, and data-driven coordination. This fragmentation results in redundant rescue efforts, delayed responses, and restricted accountability. This paper introduces StrayCare, an AI-driven, full-stack platform that consolidates rescue reporting, adoption lifecycles, and volunteer coordination into a single, cohesive framework. Empowering citizens to report cases via geotagged multimedia, the system routes data to an administrative NGO dashboard that enforces a strict state-based rescue pipeline, enabling real-time lifecycle tracking. Furthermore, the integration of an AI-powered image recognition module for species verification and an automated first-aid assistant significantly enhances triage efficiency. Engineered on a decoupled React.js and Node.js architecture, StrayCare transforms disjointed manual processes into a scalable digital framework. Initial evaluations demonstrate a significant reduction in redundant case processing and enhanced transparency throughout the adoption lifecycle.**

***Index Terms*—Stray Animal Management, Animal Adoption, Web Application, Artificial Intelligence, Geotagging, NGO Coordination, Full-Stack Architecture**

I. Introduction

The management of stray animals continues to pose profound public health and ethical challenges globally. In numerous regions, traditional population control measures rely heavily on fragmented reporting mechanisms or reactive interventions, often resulting in prolonged animal suffering and inefficient resource allocation [1]. Beyond ethical concerns, unmanaged stray populations present palpable risks, including the transmission of zoonotic diseases and localized environmental degradation [2]. These pressing realities necessitate the implementation of structured, technology-enabled interventions.

While Non-Governmental Organizations (NGOs) and civic volunteers actively partake in rescue and rehabilitation, their communication channels are frequently decentralized. Rescue operations are commonly initiated via unstructured social media announcements, which lack systematic traceability and standardized documentation [3]. Consequently, rescue efforts frequently overlap, animal medical histories remain undocumented, and prospective adopters confront disparate information.

Recent mobile and web applications—such as FurRescue [5] and the Stray Rescue Management System (SRMS) [15]—have demonstrated the feasibility of digitalizing adoption visibility and community reporting. However, a majority of these platforms fail to incorporate real-time backend coordination, robust automated verification, or intelligent analytical capabilities [7], [14].

To resolve these systemic inefficiencies, this study proposes StrayCare: a unified, full-stack digital ecosystem. By integrating geolocation-based reporting, AI-assisted image validation, and a strictly enforced case lifecycle architecture, StrayCare modernizes animal welfare operations, ensuring seamless collaboration among citizens, shelters, and administrators.

II. Related Work

The persistent requirement for improved animal welfare management has catalyzed the development of numerous technology-driven proposals. Table I provides a summary of foundational systems and their identified limitations.

The earliest iterations, such as the Stray Animal Mobile App developed in Malaysia [1], succeeded in introducing mobile incident reporting. However, these prototypes suffered from scalability constraints and the absence of concurrent multi-user coordination. Subsequent systems, including StandForPaw [3] and Animal Rescue [4], targeted community interaction and

TABLE I
LITERATURE SURVEY SUMMARY

Ref	Key Features	Key Limitations
[1]	Mobile reporting interface	Lacks multi-tier tracking and analytics
[3]	Community abuse reporting	Insufficient NGO administrative backend
[5]	AI breed detection & geofencing	Restricted exclusively to lost/found function
[10]	Location-based matching	Limited by mobile-only architecture constraints
[15]	PHP shelter management	Lacks real-time spatial case routing

basic geolocation, respectively, yet failed to integrate intelligence automation, thus imposing high manual verification burdens on administrators.

Advancements involving Artificial Intelligence were later conceptualized in FurRescue [5], which utilized TensorFlow for image detection, and in frameworks proposed by Jha et al. [7] optimizing adoption rates via machine learning. Despite their technical validity, these studies compartmentalized their solutions—focusing predominantly on the "lost-and-found" phase without bridging the gap to rehabilitation and long-term adoption. Web-based administration platforms, such as SRMS by Subramaniam and Mammi [15], established functional CRUD capabilities for shelters but lacked robust real-time API integrations and scalable geospatial rendering methodologies.

Summary of Research Gap: Despite significant strides, existing literature reveals a distinct absence of a singular ecosystem that unifies the entire continuum from initial civic report to final verified adoption. Existing platforms silo the reporting side from the administrative tracking side and omit automated AI verification integrations. StrayCare addresses this definitive gap by architecting a full-stack platform that seamlessly binds geospatial tracking, deterministic FSM (Finite State Machine) case routing, and AI-powered verifications securely within one repository.

III. PROBLEM DEFINITION

The unmitigated proliferation of stray animal populations is fundamentally exacerbated by disparate communication channels and the absence of integrated sociotechnical infrastructures. Extant systems rely predominantly on ad-hoc coordination tools or loosely structured applications, cultivating severe operational inefficiencies. Consequently, the core problem addressed is the profound absence of a unified, AI-supported digital environment that facilitates transparent and real-time collaboration among citizens, NGOs, and governing authorities. The StrayCare framework is meticulously designed to dismantle this inefficiency by establishing a cloud-native application integrating rescue mechanics, geolocation validation, and immutable lifecycle tracking.

IV. METHODOLOGY AND SYSTEM ARCHITECTURE

A. Proposed Approach

StrayCare departs from fragmented operational models by situating all stakeholders—citizens and NGOs—within a centralized, cloud-based platform. The system enforces an unbroken chain of custody for every animal reported. Utilizing a three-tier modular architecture, StrayCare comprises:

1) **Presentation Layer (React.js):** Interfaces seamlessly with Google Maps APIs to generate responsive, geotagged reporting interfaces.
2) **Application Layer (Node.js/Express.js APIs):** Executes complex case routing, Finite State Machine logic transitions, and enforces security via robust JSON Web Token (JWT) authorizations.
3) **Data Layer (MongoDB & Cloud Object Storage):** Facilitates dynamic NoSQL operations for rapid multi-dimensional querying, coupled with secure Firebase/Cloudinary cloud buckets for media persistence.

B. Functional Modules and Workflow

1) **User Module:** Allows citizens to report animals by uploading photographs and capturing exact GPS coordinates.
2) **NGO Dashboard Module:** Serving as the administrative nexus, this module permits verified NGOs to spatially visualize open reports, triage rescues, and append veterinary logs.
3) **AI Chatbot & Verification Module:** Implements an automated LLM-powered assistant for immediate civic first-aid guidance. A complementary image model evaluates uploads for species verification before writing to the database.
4) **Adoption Management Module:** Upon completion of rehabilitation, animals seamlessly transition from an "Under Care" state to "Available for Adoption". The complete rescue history remains visibly linked to the adoption profile, guaranteeing total transparency for potential adopters.

V. IMPLEMENTATION AND ANALYSIS

StrayCare was implemented utilizing an API-driven decoupled approach. The frontend React single-page application interacts exclusively via RESTful principles with the Node/Express backend.

A. Originality and Technical Depth

StrayCare distinguishes itself conceptually and architecturally from platforms evaluated in the literature review through the following contributions:

- **Lifecycle State Enforcement:** StrayCare implements a rigorous state machine for cases: Reported $\rightarrow$ Verified $\rightarrow$ Under Care $\rightarrow$ Available for Adoption $\rightarrow$ Adopted. Unlike architectures in previous applications [5], [15], role-permissible actions are restricted contextually at the backend.
- **Geospatial and Analytical Integration:** Google Maps API enables instantaneous reverse geocoding and density visualization. By linking GPS matrices directly horizontally with MongoDB records, NGOs can spatially optimize their dispatch vectors.

- **Full-Spectrum Traceability:** The entity relationship schema robustly links Users, Cases, and Adoptable Pets. The adoption marketplace is not an isolated database table; it inherits the entire medical and situational history of the initial rescue, establishing definitive operational trust.

VI. Results & Evaluation

To evaluate structural efficiency, workflow simulations and localized testing were conducted juxtaposing StrayCare against decentralized reporting methods.

A. *Operational Efficiency Metrics*

1) **Reporting Redundancy Reduction:** Through the utilization of spatial mapping and image-verification logic, duplicated reports of the same animal within a specific coordinate radius were identified pre-emptively on the NGO dashboard. This mechanism yielded a significant reduction in report redundancy, filtering out visual noise.
2) **Case Processing Velocity:** Centralizing communication through the NGO Dashboard dramatically lowered the overhead time required to transition a raw citizen report into an active "Under Care" status.

B. *Comparative Analysis*

Table II highlights the functional advantages of StrayCare against identified gaps in traditional systems.

TABLE II
Functional Completeness Comparison

Identified Gap	StrayCare Implementation
Fragmented social media reporting [16]	Authenticated, geotagged "Report Case" formulation
Opaque rescue status visibility [6]	Citizen UI exhibiting deterministic case state updates
Disconnected adoption workflows [5]	Linked lifecycle preserving rescue and medical logs
Unverified case overhead [1]	Integrated AI image tracking and secure triage routing

VII. Conclusion

StrayCare establishes a highly integrated, data-backed digital framework intended to systematically rectify the inefficiencies plaguing stray animal welfare operations. By migrating disjointed reporting behaviors onto a rigorous, map-based, state-enforced architecture, the platform significantly diminishes administrative redundancy while elevating public transparency. While its current iteration relies upon active network connectivity, the foundational architecture successfully proves that consolidating advanced AI modules, geospatial analytics, and robust full-stack engineering yields a powerful and sustainable ecosystem for modernizing civic animal rescue.

VIII. Future Scope

1) **Android Application Integration:** Utilizing cross-platform frameworks to deploy offline-first mobile reporting tools.
2) **Predictive AI for Adoptions:** Deepening machine learning integrations to predict and match shelter dog temperaments with user lifestyle compatibility metrics prior to adoption.
3) **Veterinary API Federation:** Generating interoperable data pipelines with registered private clinics for automated syncing of clinical pet records directly into the NGO dashboard.

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